



B.Tech. Degree VII Semester Regular/Supplementary Examination November 2024

ME 19-205-0702 VIBRATION AND NOISE CONTROL (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Model a given physical system using spring, mass and damping elements.
 CO2: Formulate the governing differential equations of single dof, multi dof and continuous systems.
 CO3: Solve the governing equations and interpret the results.
 CO4: Understand the effects of noise and the control measures adopted.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

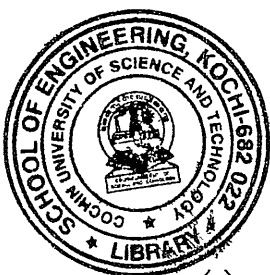
PART A

(Answer ALL questions)

(8 × 3 = 24)

- | | Marks | BL | CO | PO |
|---|-------|----|----|--|
| I. (a) Determine the equivalent spring constant of the system shown, if the springs have a stiffness of $k=1\text{ kN/m}$. | 3 | L2 | 1 | 1,2,3 |
| | | | | |
| (b) Match the following for a single degree of freedom system with $m = 1\text{ kg}$, $k = 2\text{ N/m}$, $c = 0.5\text{ Ns/m}$. | 3 | L3 | 1 | 1,2,3 |
| 1. Natural frequency (rad/s), ω_n
2. Natural time period (s), τ_n
3. Damping ratio, ζ
4. Critical damping coefficient, (Ns/m), c_c
5. Damped frequency (rad/s), ω_d
6. Logarithmic decrement, δ | | | | a. 1.3919
b. 2.8284
c. 1.1285
d. 1.4142
e. 0.1768
f. 4.4429 |
| (c) A spring-mass-damper system, having an undamped natural frequency of 100 Hz and a damping constant of 20 Ns/m , is used as an accelerometer to measure the vibration of a machine operating at a speed of 3000 rpm . If the actual acceleration is 10 m/s^2 and the recorded acceleration is 9 m/s^2 , find the mass and spring constant of the accelerometer. | 3 | L2 | 3 | 1,2,3 |
| (d) What are the conditions to be satisfied for converting a torsional geared system into an equivalent three rotor system? | 3 | L1 | 1 | 1,2,3 |
| (e) Derive an expression for the additional deflection 'y' of a shaft with an eccentric mass 'm' at a distance 'e' from its axis of rotation, having an angular velocity of $\omega\text{ rad/s}$. | 3 | L2 | 2 | 1,2,3 |
| (f) Derive the differential equation of motion for free axial vibration of a rod. | 3 | L2 | 2 | 1,2,3 |
| (g) Differentiate between audible sound, infrasound and ultrasound. | 3 | L1 | 4 | 1,2 |
| (h) What are the methods adopted for the control of noise at the receiver? | 3 | L1 | 4 | 1,2 |

(P.T.O.)

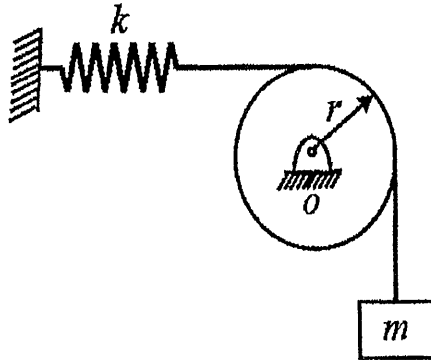


PART B

(4 × 12 = 48)

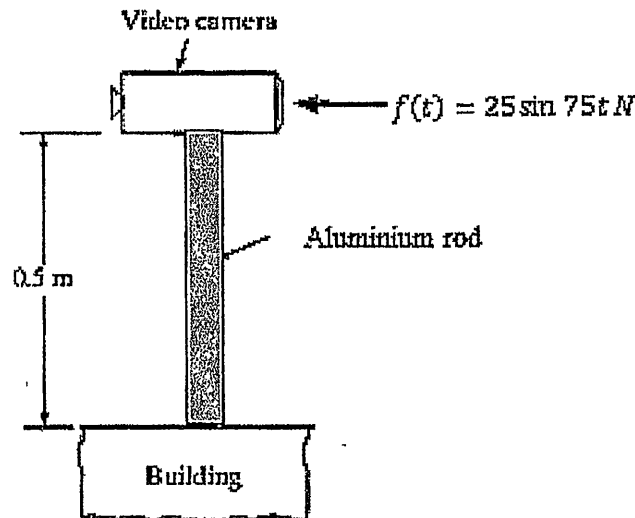
Marks	BL	CO	PO
4	L3	1	1,2,3

- II. (a) A block of mass m is connected to a spring of stiffness k through a cable, which passes through a pulley as shown in figure. The pulley rotates about a fixed center O . The mass moment of inertia of the pulley about its centre of gravity is I_o . Determine the natural frequency of vibration of the system.



- (b) A video camera, of mass 2 kg , is mounted on the top of a bank building for surveillance. The video camera is fitted at one end of an aluminium rod whose other end is fixed to the building as shown. The wind induced force acting on the video camera is $f(t) = 25 \sin(75t)\text{ N}$.

Model the system as a single degree of freedom undamped spring-mass system. Write down the governing differential equation of the system. What is the diameter of the aluminium rod if the maximum amplitude of vibration of the video camera is to be limited to 0.005 m . Take the modulus of elasticity of aluminium $E = 80\text{ GPa}$.



OR

(Continued)

Marks	BL	CO	PO
4	L3	1	1,2,3

- III. (a) Model the given system as a single degree of freedom spring-mass system with output response x of mass m_2 , if

$$k_1 = 1 \text{ kN/m},$$

$$k_2 = 2 \text{ kN/m},$$

$$k_t = 3 \text{ kNm/rad}$$

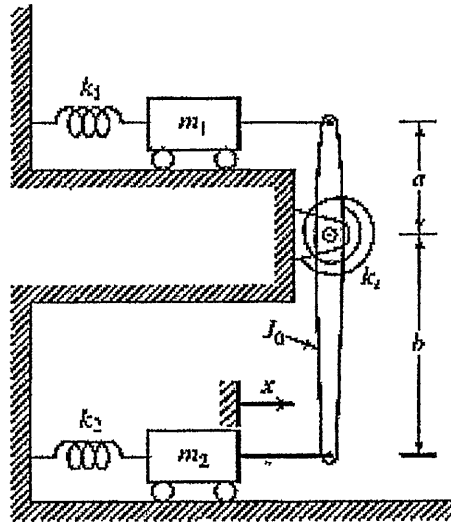
$$m_1 = 5 \text{ kg},$$

$$m_2 = 10 \text{ kg},$$

$$J_o = 15 \text{ kg.m}^2$$

$$a = 1 \text{ m},$$

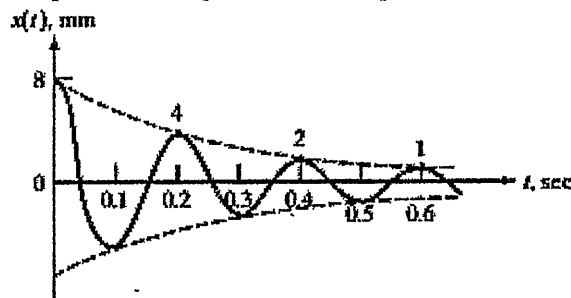
$$b = 2 \text{ m}.$$



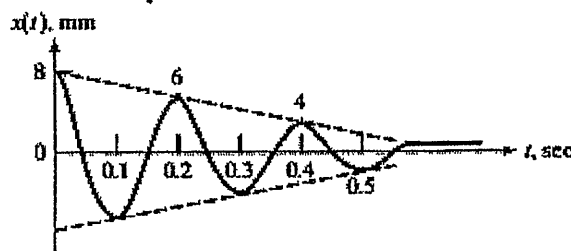
- (b) The free vibration response of an electric motor of weight 500 N mounted on two different types of foundations are shown. Identify the nature of damping provided by the foundation in each case. Justify your answer. Also determine

8	L4	3	1,2,3
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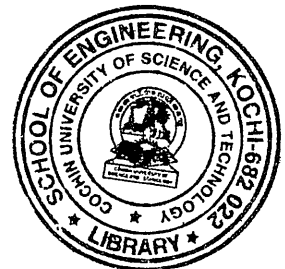
- the spring constant and damping coefficient of the foundation in each case.
- undamped and damped natural frequencies in each case.



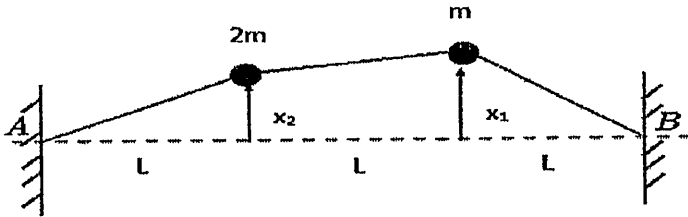
(a)



(b)



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		Marks	BL	CO	PO
IV.	Determine the natural frequencies and mode shapes of the vibrating system as shown in figure. Assume that the tension T in the string remains unchanged for small angles of oscillation. Take $m = 1$ kg, $L = 0.25$ m and $T = 25$ N.	12	L4	3	1,2,3
					
OR					
V.	A shaft 50 cm long is attached with two rotors A and B at the opposite ends. The mass of the rotor A is 30 kg and its radius of gyration is 30 cm and the corresponding values of rotor B are 50 kg and 40 cm respectively. The shaft is 7 cm in diameter for the first 25 cm, 12 cm in diameter for the next 10 cm and 10 cm in diameter for the remainder of its length. Modulus of rigidity for the shaft material is 80 GN/m ² . Determine (i) the frequency of torsional vibration. (ii) the position of the node. (iii) the amplitude ratio.	12	L4	3	1,2,3
VI.	A shaft of diameter 50 mm and length 3 m is mounted on short bearings at the ends and carries two masses of 100 kg and 200 kg at 1 m and 2 m respectively from the left support. The modulus of elasticity for the shaft material is 200 GN/m ² . Determine the frequency of transverse vibration using (i) Dunkerley's method. (ii) Rayleigh's method. Also determine the critical speed of the shaft.	12	L4	3	1,2,3
OR					
VII	A uniform beam fixed at one end and free at the other is vibrating freely in the transverse direction. Derive the frequency equation.	12	L3	2	1,2,3
VIII. (a)	What is an octave band? Differentiate between full and one-third octave bands.	4	L1	4	1,2
(b)	Derive an expression for sound power level in terms of sound intensity level and acoustic surface area.	8	L2	4	1,2
OR					
IX. (a)	Differentiate between inverse law and inverse square law with respect to distance from a source as applied to acoustic waves.	4	L1	4	1,2
(b)	When operating independently, three machines create a noise level of 90 dB, 94 dB and 98 dB. Determine the overall sound pressure level and average sound pressure level when they operate simultaneously.	8	L3	4	1,2

Bloom's Taxonomy Level's

L1 – 14.2%, L2 – 16.7%, L3 – 25.8%, L4 – 43.3%.
